MOBILE PHONE PRICE USING MACHINE LEARNING

Internship Project Presentation

Organization: NextHikes IT Solutions

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INTRODUCTION TO THE PROJECT

- In today's smartphone market, the price of a device depends on many technical specifications such as RAM, processor, storage, battery capacity, and camera features.
- This project focuses on using **machine learning techniques** to predict the price of mobile phones based on their specifications.
- By analyzing the dataset, we can identify which features have the most influence on mobile phone pricing.

PROJECT OBJECTIVE

The main goal of this project is to build a system that can estimate the price of a smartphone using its specifications.

- Objectives of this project:
  - Analyze the smartphone dataset
  - Identify the key factors affecting phone prices
  - Build machine learning models for prediction
  - Compare different models to find the most accurate one

PROBLEM STATEMENT

- Mobile phone prices vary widely depending on hardware and technical specifications.
- For smartphone companies and sellers, it is important to understand which features affect pricing the most.
- The aim of this project is to develop a predictive model that can estimate the price of a mobile phone based on its specifications.

DATASET DESCRIPTION

The dataset used in this project contains information about different smartphone models and their specifications.

- Some of the main attributes in the dataset include:
  - Model Name
  - Colour
  - RAM
  - Internal Storage
  - Battery Capacity
  - Processor
  - Rear Camera
  - Front Camera
  - Mobile Height
- The **Price** column is used as the **target variable** for prediction.

DATA EXPLORATION

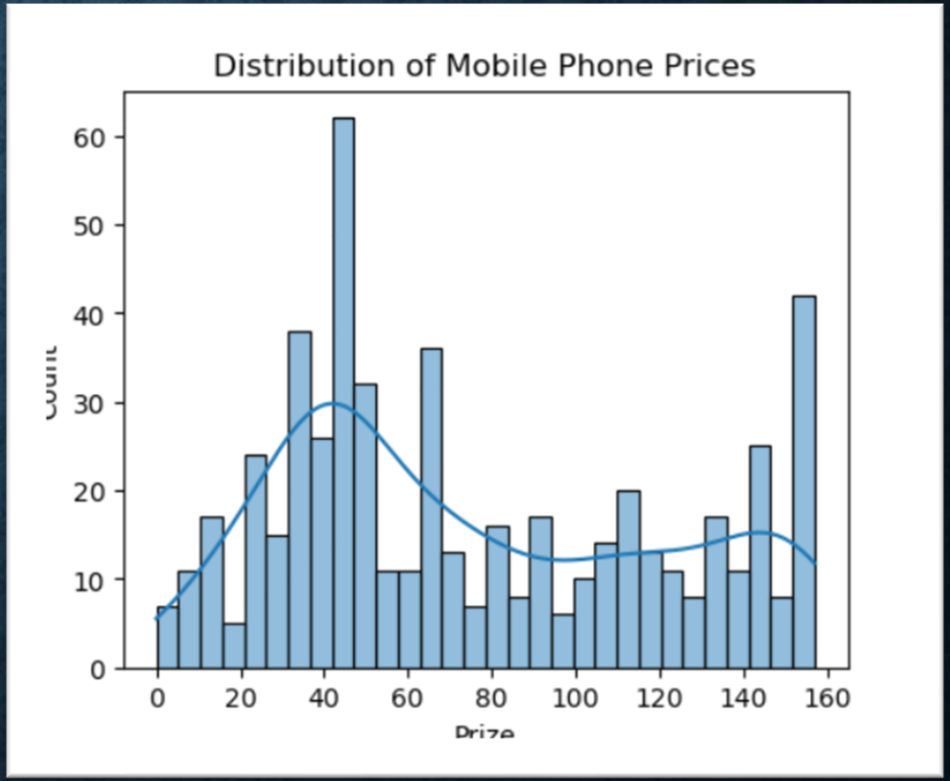
- Before building the machine learning model, the dataset was explored to understand its structure.
- The following steps were performed:
  - Checked the number of rows and columns
  - Examined data types of each feature
  - Analyzed statistical summaries
  - Checked for missing values
- This step helps in understanding the dataset and preparing it for further analysis.

DATA PREPROCESSING

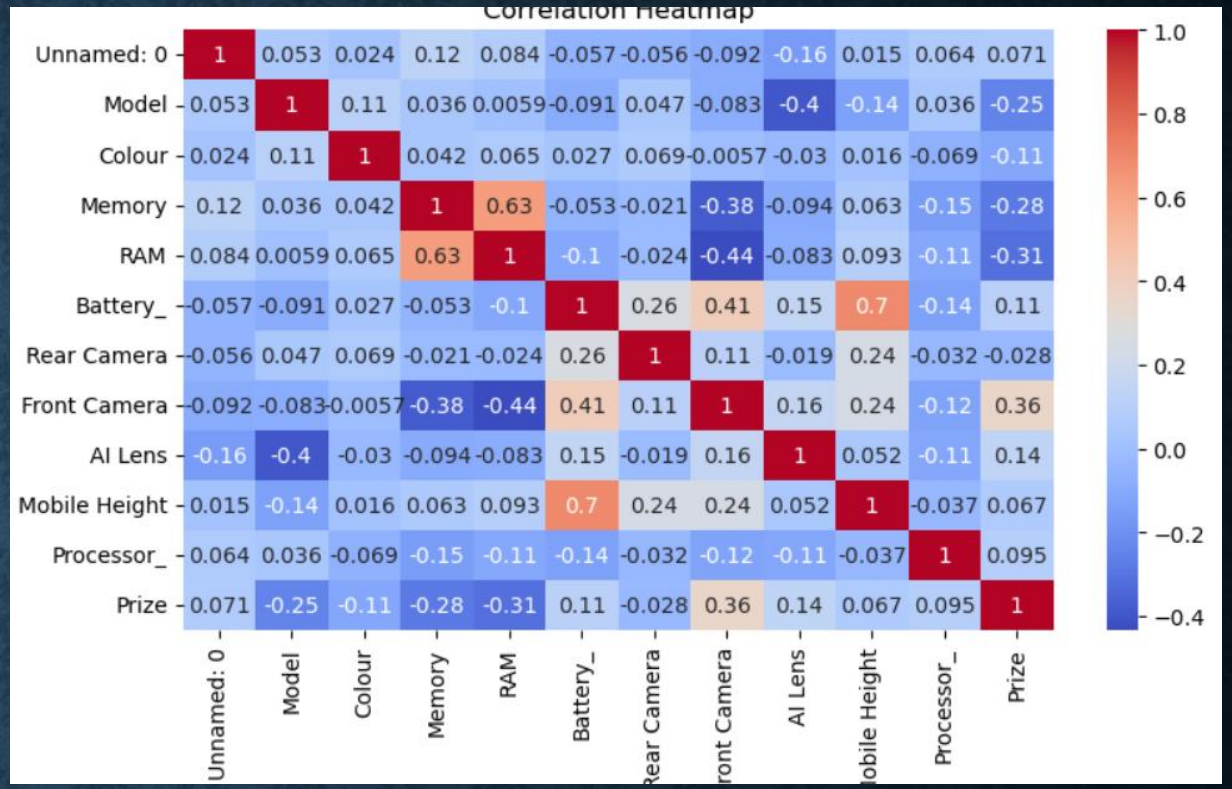
- Data preprocessing is an important step before training machine learning models.
- The following preprocessing steps were applied:
- Handling missing values in the dataset
- Encoding categorical features into numerical form
- Preparing the dataset for model training
- Proper preprocessing improves the performance of machine learning algorithms.

PRICE DISTRIBUTION ANALYSIS

- The price distribution graph shows how the prices of mobile phones are spread across the dataset.
- This visualization helps us understand whether most phones belong to a specific price range or if prices are evenly distributed.
- It also helps identify patterns or unusual values in pricing.



CORRELATION ANALYSIS



- A correlation heatmap was used to understand the relationship between different features in the dataset.
- Features that show a strong relationship with the price variable are considered important for prediction.
- This step helps identify which attributes are likely to influence the model's predictions.

IMPORTANT FEATURES

Feature analysis helped identify the specifications that have the greatest impact on mobile phone prices.

- Some key features that influence price include:
  - RAM capacity
  - Processor type
  - Battery capacity
  - Internal storage
  - Camera specifications
- These factors are commonly associated with device performance and quality.

MODEL DEVELOPMENT

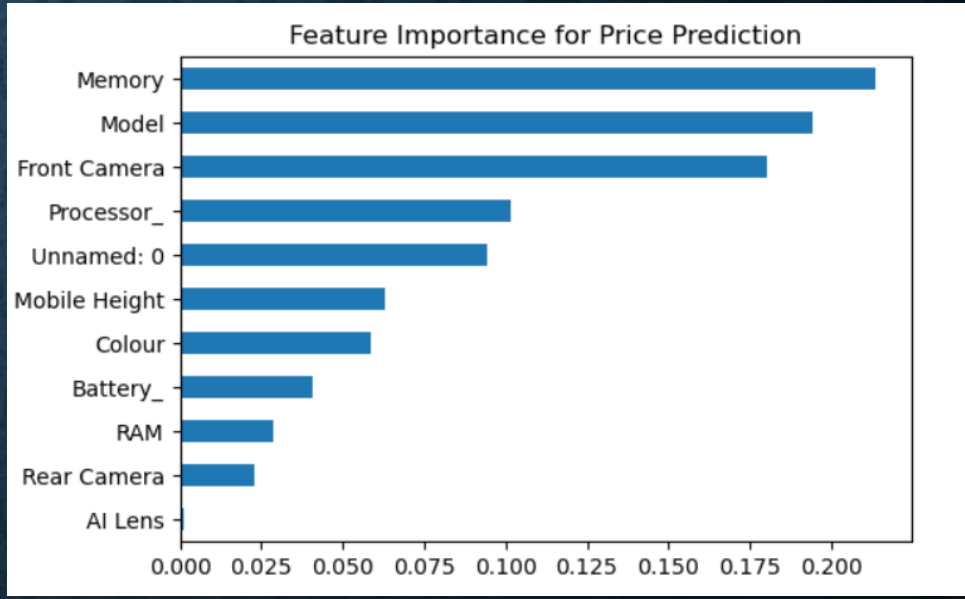
- Different machine learning models were implemented to predict mobile phone prices.
- The models used in this project include:
- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Each model was trained using the training dataset and then tested on unseen data.

MODEL EVALUATION

- To evaluate the performance of the models, several evaluation metrics were used.
- These include:
- **MAE (Mean Absolute Error)** – measures average prediction error
- **RMSE (Root Mean Squared Error)** – penalizes larger errors
- **R^2 Score** – indicates how well the model explains the data
- The models were compared based on these metrics to identify the most accurate one.

FEATURE IMPORTANCE AND INSIGHTS

The feature importance graph highlights which smartphone specifications have the strongest influence on price predictions.



- Key insights from the analysis include:
  - Higher RAM generally increases phone price
  - Powerful processors are associated with premium devices
  - Larger battery capacity affects mid-range phone pricing
  - Advanced camera features contribute to higher prices
- These insights can help companies understand pricing strategies.

CONCLUSION

- In this project, machine learning techniques were used to predict mobile phone prices based on their specifications. The dataset was analyzed to understand how different features affect smartphone pricing. Models such as Linear Regression, Decision Tree, and Random Forest were applied and compared. The analysis showed that features like RAM, processor, and storage have a strong impact on mobile phone prices.

Thankyou